Active sensing explains cortical variability and improves visual information

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What looks like noise may reflect active sensing

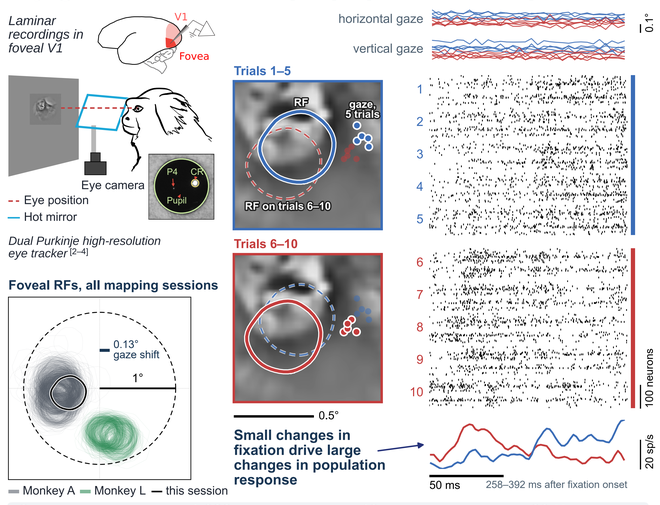
Neurons respond differently across repeated presentations of the same stimulus. These trial-to-trial differences are often described as "noise," but they can include sensory effects of the animal's own movements. Even during fixation, small eye movements continually reshape visual input.

How much cortical variability comes from the eye?

Does that motion help encode spatial detail?

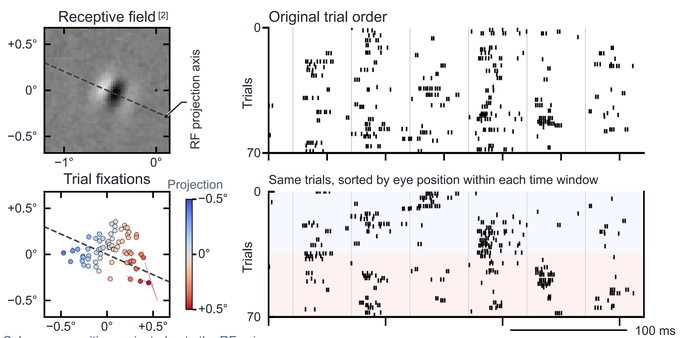
We combine precise eye tracking, recordings from primary visual cortex (V1), and a model fitted to neural responses to address these questions.

1 Tiny gaze shifts reshape population responses



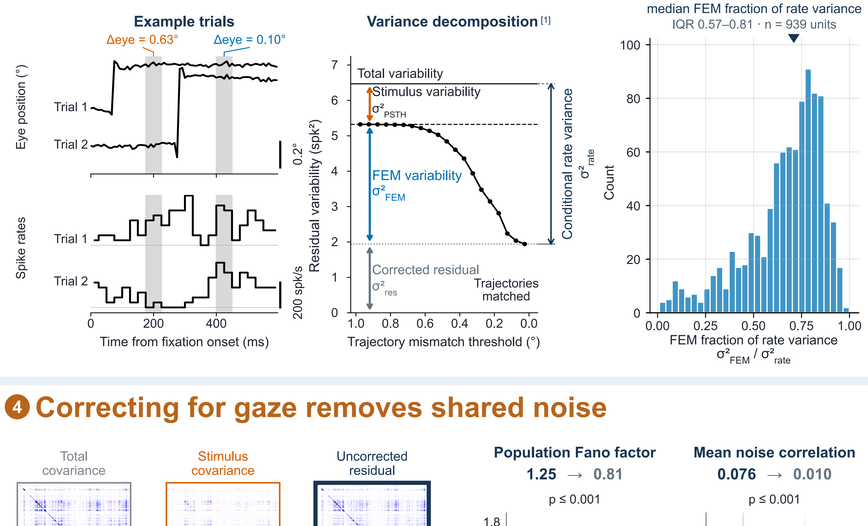
2	1,022	19	20 Hz
awake marmosets	foveal V1 units	recording sessions	repeated natural images

2 One neuron's "noise" tracks the eye



Colors: eye position projected onto the RF axis.

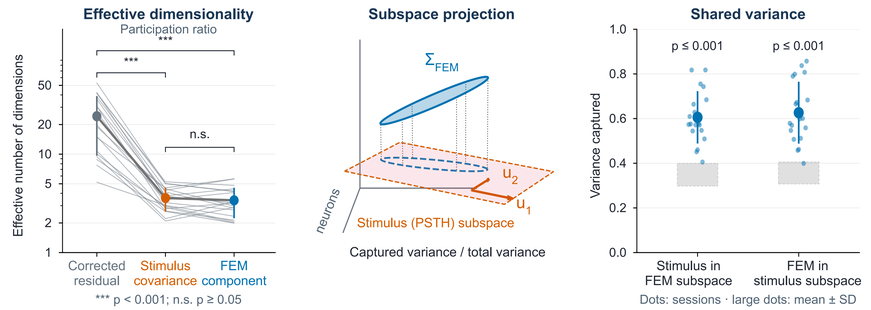
3 Eye movements drive most single-cell rate variance (71%)



4 Correcting for gaze removes shared noise

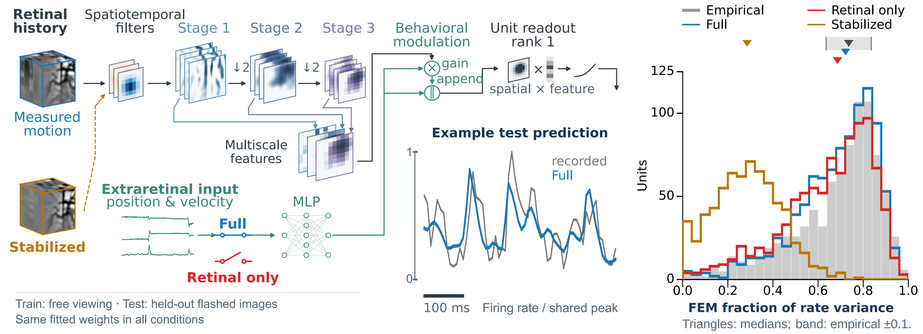
19 sessions; 25-ms bins. Points: mean ± SD. Gray bands: 95% shuffle null; violins: pairwise correlations.
Corrected mean noise correlation: 95% CI −0.010 to 0.029. Both tests compare correction with shuffle.

5 Eye-driven variability follows stimulus dimensions



*** p < 0.001; n.s. p ≥ 0.05
25-ms counts; alignment uses four leading dimensions per component.
Dots: sessions · large dots: mean ± SD
Gray bands: 95% shuffle-null intervals

6 Gaze-dependent retinal input predicts single trials

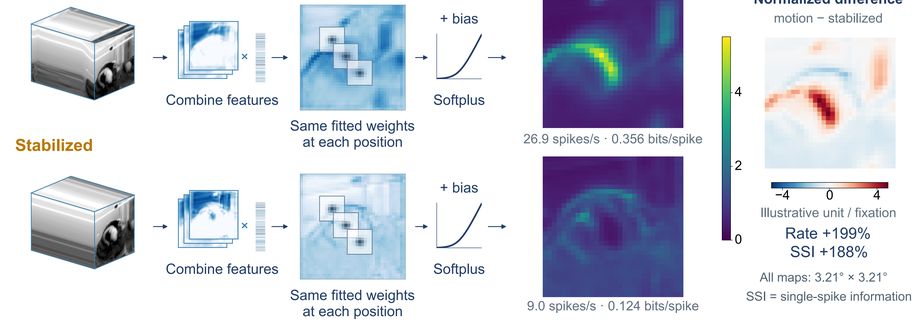


Train: free viewing · Test: held-out flashed images
Same fitted weights in all conditions

Triangles: medians; band: empirical ±0.1.

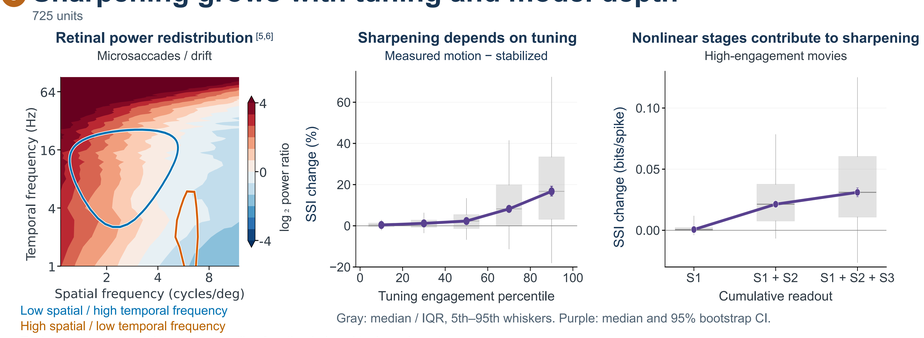
Median fraction of conditional rate variance explained 962 units · 19 sessions	Trial average	Full	Retinal only	Stabilized
	0.139	0.235	0.226	0.042

7 In the model, eye motion sharpens spatial coding



8 Sharpening grows with tuning and model depth

725 units



Low spatial / high temporal frequency
High spatial / low temporal frequency

Tuning engagement: within-unit percentile of motion-induced tuning-relevant power.
All quintiles/stages > 0 (Holm-adjusted p < 0.05).

Gray: median / IQR, 5th–95th whiskers. Purple: median and 95% bootstrap CI.

References
[1] McFarland 2016 · [2] Yates 2023 · [3] Wu 2023
[4] Ressmeyer 2026 · [5] Rucci & Victor 2015 · [6] Kuang 2012

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